

Lab Report 3: Triplication and SPARQL

Course: II.3521 – Web Semantics and Knowledge Management

Exercise: Exercise 4 – Loading and Inspecting the RDF Graph

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Task 1: Verify the Dataset:

After loading the integrated RDF dataset, Academic_KG.ttl, into an RDF triplestore, the following information was recorded:

- a) Total triples loaded: 45 triples (as reported in our team's lab report).
- b) Classes represented: Course, Professor, Skill, Program and CareerPath.
- c) Sample resources: CS301, P01, S01, MasterDataScience and Data Scientist.
- d) Sample relationships: isTaughtBy, develops Skill, has Prerequisite, requires Skill and isPartOf.

The screenshot shows the Apache Jena Fuseki web interface at the URL `localhost:3030/#/dataset/academic/upload`. The page title is `/academic`. Below the navigation bar, there are tabs for `query`, `add data`, `edit`, and `info`. The main section is titled `Upload files /academic/data` with a subtext: "Load data into the default graph of the currently selected dataset, or the given named graph. You may upload any RDF format, such as Turtle, RDF/XML or TRIG." Below this, there is a text input field for "Dataset graph name" with the placeholder "Leave blank for default graph". Under the "Files to upload" section, there are two buttons: `+ select files` and `upload all`. A table displays the upload progress for `Academic_KG.ttl`. The table has columns: `name`, `size`, `speed`, `status`, and `actions`. The row for `Academic_KG.ttl` shows a size of `1.62kb`, a speed of `1.82kb/s`, and a status of `100.00` with a green progress bar. Below the status bar, it says "Triples uploaded: 45". The `actions` column contains two buttons: `upload now` and `remove`.

name	size	speed	status	actions
Academic_KG.ttl	1.62kb	1.82kb/s	100.00 Triples uploaded: 45	<button>upload now</button> <button>remove</button>

Figure 1: Successful loading of Academic_KG.ttl in Apache Jena Fuseki, showing 45 triples.

localhost:3030/#/dataset/academic/query

```

1 PREFIX ex: <http://example.org/academic/>
2 PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
3
4 SELECT ?class (COUNT(?entity) AS ?nbEntities)
5 WHERE {
6   ?entity rdf:type ?class .
7 }
8 GROUP BY ?class

```

Table Response Geo 5 results in 0.094 seconds Simple view Ellipse Filter query results Page size: 50

class	nbEntities
1<http://example.org/academic/AcademicProgram>	"1"^^<http://www.w3.org/2001/XMLSchema#integer>
2<http://example.org/academic/Skill>	"4"^^<http://www.w3.org/2001/XMLSchema#integer>
3<http://example.org/academic/CareerPath>	"1"^^<http://www.w3.org/2001/XMLSchema#integer>
4<http://example.org/academic/Course>	"3"^^<http://www.w3.org/2001/XMLSchema#integer>
5<http://example.org/academic/Professor>	"3"^^<http://www.w3.org/2001/XMLSchema#integer>

Showing 1 to 5 of 5 entries

Figure 2: Verification of RDF classes and entity counts using a SPARQL query in Apache Jena Fuseki.

Task 2: Inspect the Graph:

localhost:3030/#/dataset/academic/query

Table Response Geo 10 results in 0.055 seconds Simple view Ellipse Filter query results Page size: 50

s	p	o
1<http://example.org/academic/P01>	<http://example.org/academic/name>	Anna Mueller
2<http://example.org/academic/P01>	<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>	<http://example.org/academic/Professor>
3<http://example.org/academic/P01>	<http://example.org/academic/teaches>	<http://example.org/academic/CS301>
4<http://example.org/academic/DataScientist>	<http://example.org/academic/name>	Data Scientist
5<http://example.org/academic/DataScientist>	<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>	<http://example.org/academic/CareerPath>
6<http://example.org/academic/DataScientist>	<http://example.org/academic/requiresSkill>	<http://example.org/academic/S01>
7<http://example.org/academic/DataScientist>	<http://example.org/academic/requiresSkill>	<http://example.org/academic/S02>
8<http://example.org/academic/DataScientist>	<http://example.org/academic/requiresSkill>	<http://example.org/academic/S03>
9<http://example.org/academic/S01>	<http://example.org/academic/name>	Machine Learning
10<http://example.org/academic/S01>	<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>	<http://example.org/academic/Skill>

Showing 1 to 10 of 10 entries

Figure 3: Inspection of RDF triples and relationships using a SPARQL query in Apache Jena Fuseki.

The RDF graph was inspected to understand how courses, professors, skills, academic programs and career paths are connected.

Question: Why can a graph representation make some relationships easier to explore than a traditional relational representation?

In a relational database, finding connected information across multiple tables may require complex JOIN queries. In an RDF graph, relationships are represented as direct links between entities. This makes it easier to follow connections between courses, skills, professors and career paths using SPARQL queries.